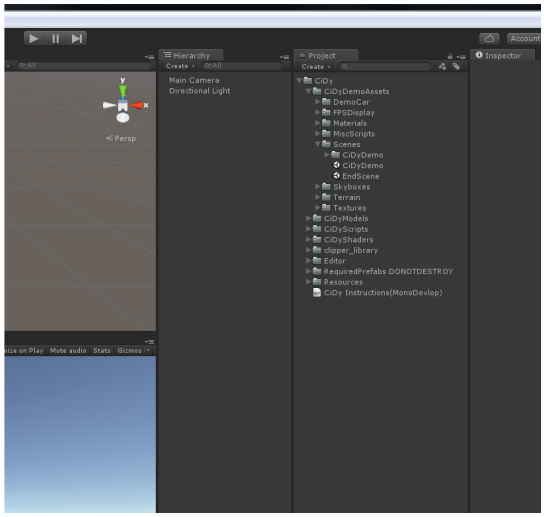
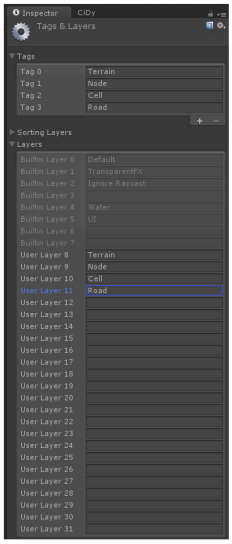
CiDy Controls and Quick Guide.

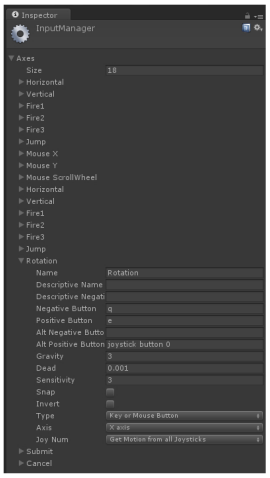
Once CiDy is Unpacked into the Unity Project you will have the CiDy Folder set in your project window.



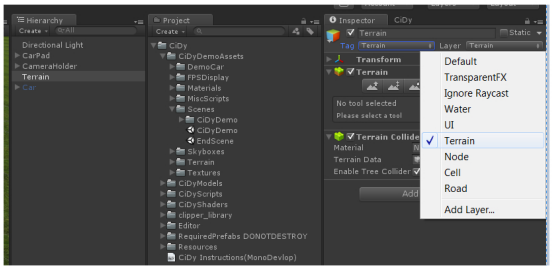
Before we can use the CiDy Editor. We must add these Strings to the Tags/Layer Manager in Unity. "Terrain","Node","Cell","Road".



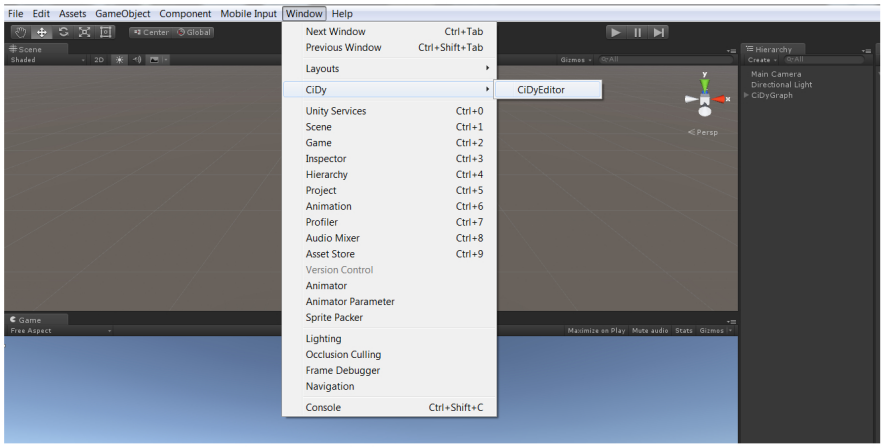
To Use the Camera in the CiDy demo scene we need to add a new Key Input in the Input Manager.
Enter "Rotation" as a Key Name.
Default Negative Input = q.
Positive Input = e.
Gravity = 3.
Sensitivity = 3.



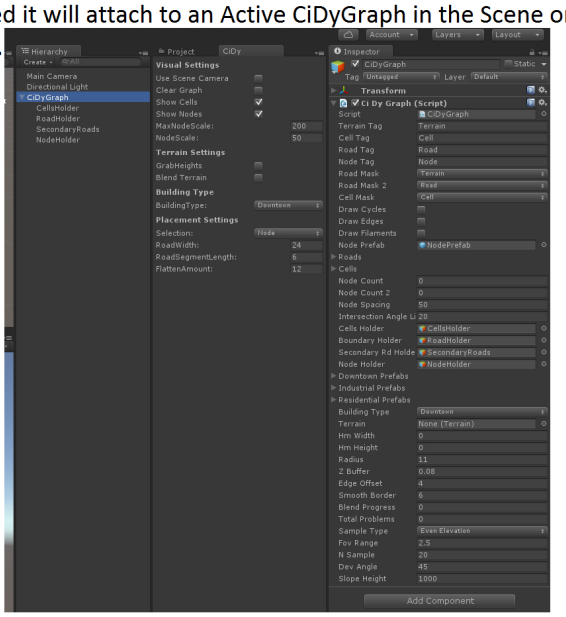
For CiDy to build on any surface with a Collider Attached. That Objects tag/Layer must be set using the Terrain Tag/Layer we added.



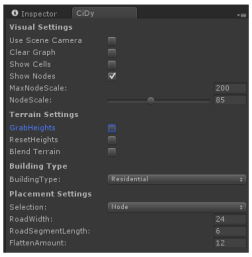
Now that the Inputs are setup we can open the CiDyEditor Window and begin developing.
Open the CiDyEditor Window in the Unity Tool Bar Window Drop down. Window-CiDy-CiDyEditor.



Everytime the Editor Window is created it will attach to an Active CiDyGraph in the Scene or create a new one if one does not exist.



Now lets go over the CiDyEditor Window settings.

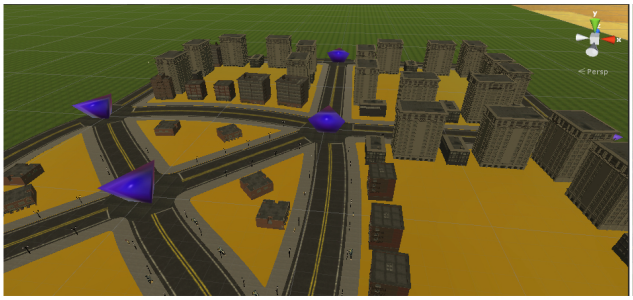


UseSceneCamera : Used to switch between scene placment control and standard scene control.

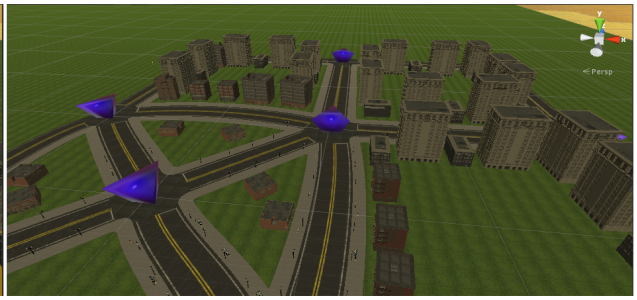
Clear Graph : Used to Clear the CiDyGraph of all created and placed meshes.

Before Clear Graph **After Clear Graph**

Show Cells : Used to enable/disable Cell Colliders.(Required Active if you want to modify a cell)

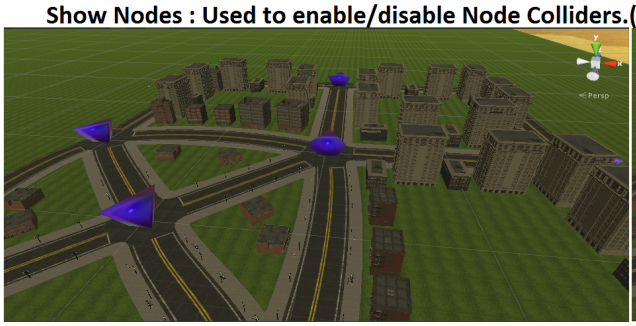


Cells On

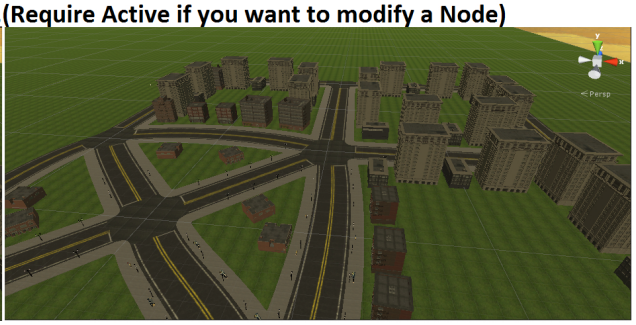


Cells Off

Show Nodes : Used to enable/disable Node Colliders.(Require Active if you want to modify a Node)

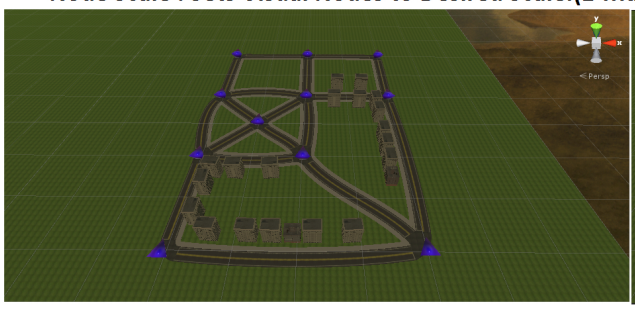


Nodes On

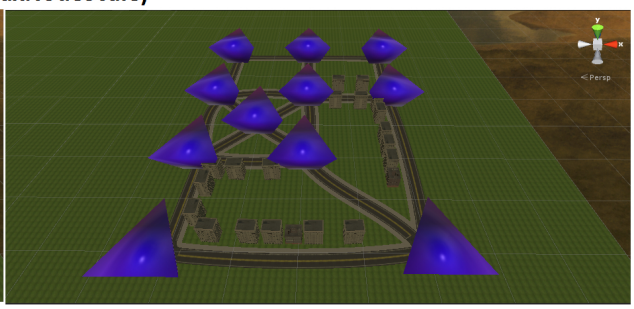


Nodes Off

Max Node Scale : Adjust max node scale.
Node Scale : Sets Visual Nodes to Desired Scale.(1-MaxNodeScale)



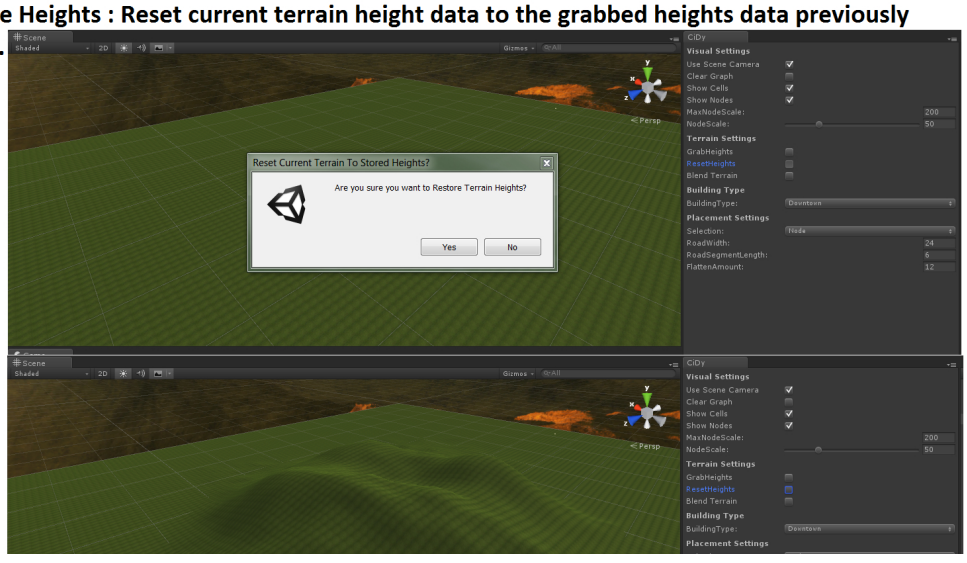
Node Scale = 50



Node Scale = 200

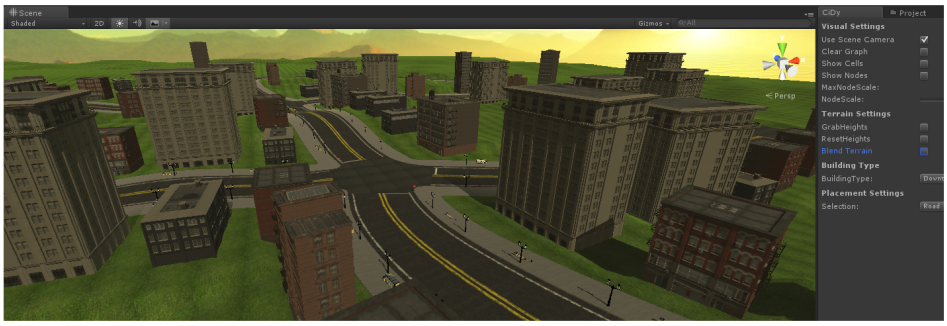
Now lets look at Terrain Settings.
Grab Heights : This will store the current terrain height data into the CiDyGraph.

Restore Heights : Reset current terrain height data to the grabbed heights data previously stored.



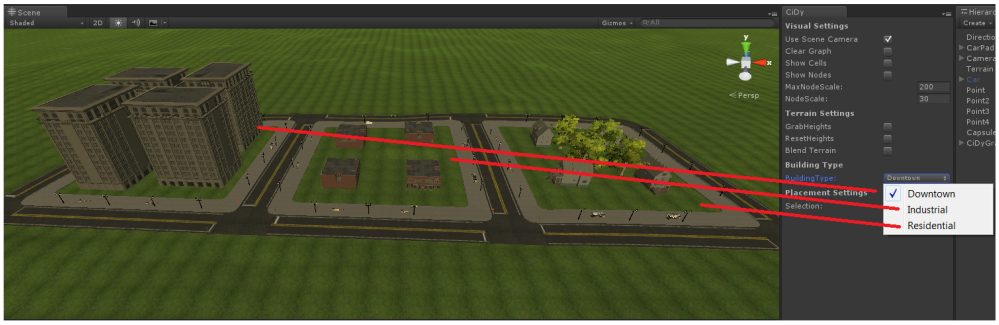
Blend Terrain : This will blend the terrain to the CiDyGraph assets.(Recommend backing up all terrain data before blending)

Before Blend



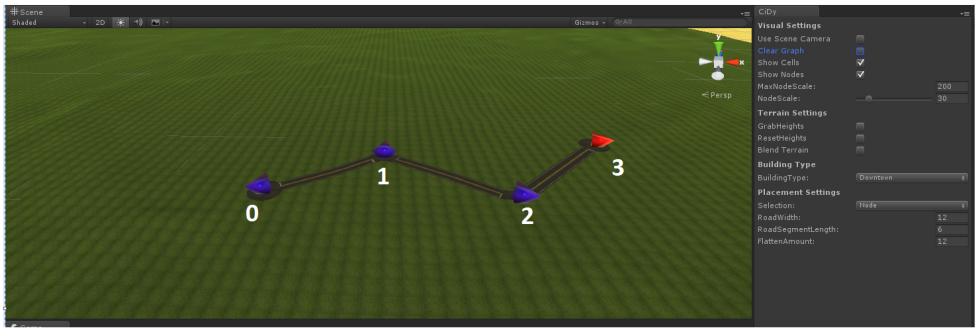
After Blend

Building Type Enum : This is used to pre determine building type for newly created cells.



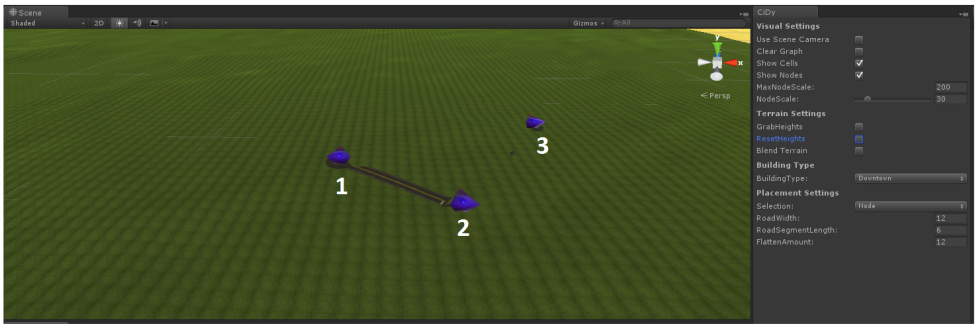
Now lets see the Placement Settings. These change based on Selection Enum Type.

Selection Node : Right Mouse Button will create new Nodes or connect two nodes if one is selected.

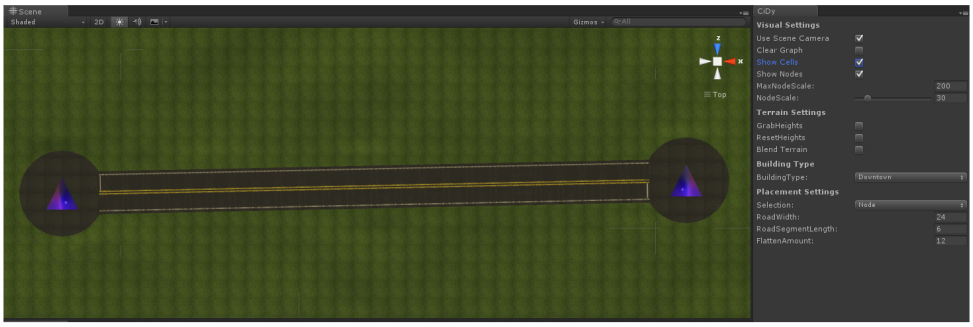


Chained Nodes in Editor

Control plus Left Mouse Button will Destroy Nodes. Cntrl+LMB



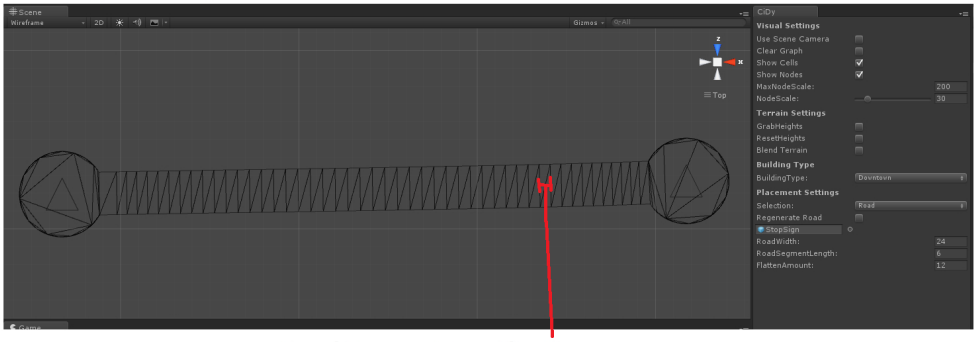
Road Width : Determines the Road Mesh Width.



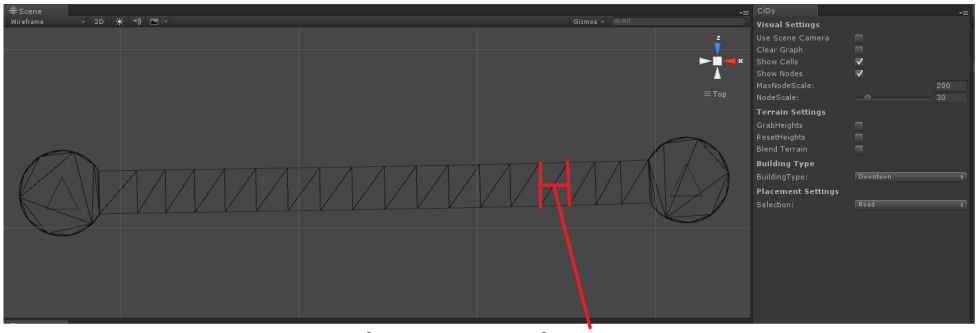
Road Width = 24

Road Width = 12

Road Segment Length : Determines distance between road mesh segments.

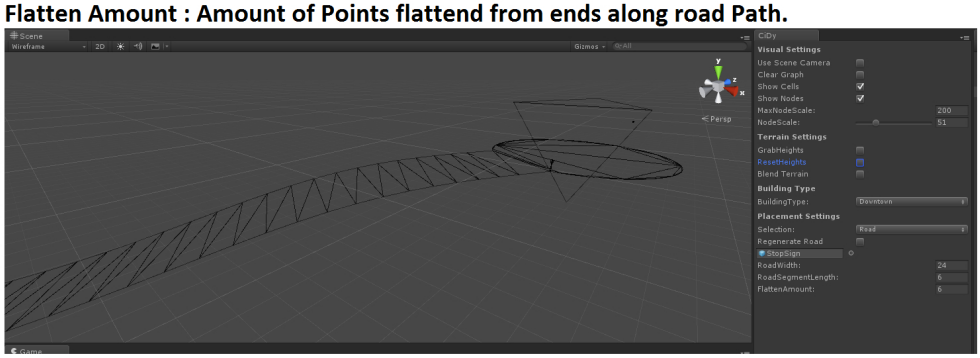


Road Segment Length = 6

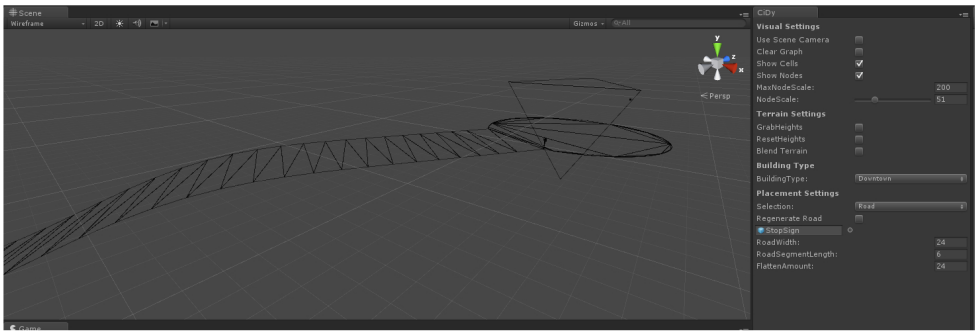


Road Segment Length = 12

Flatten Amount : Amount of Points flattend from ends along road Path.

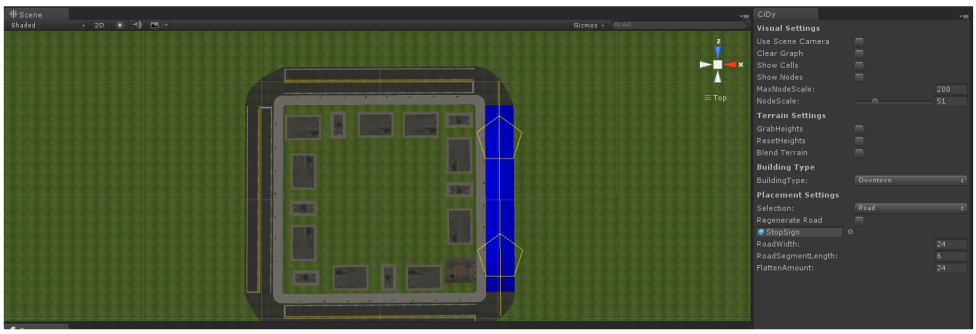


Flatten Amount = 6

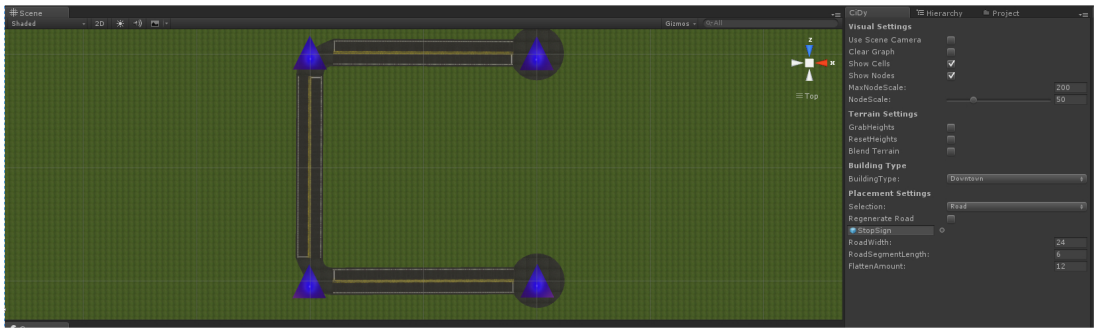


Flatten Amount = 24

Now lets look at Road Selection.
Left Mouse Button is used to Select/Deselect a Road.



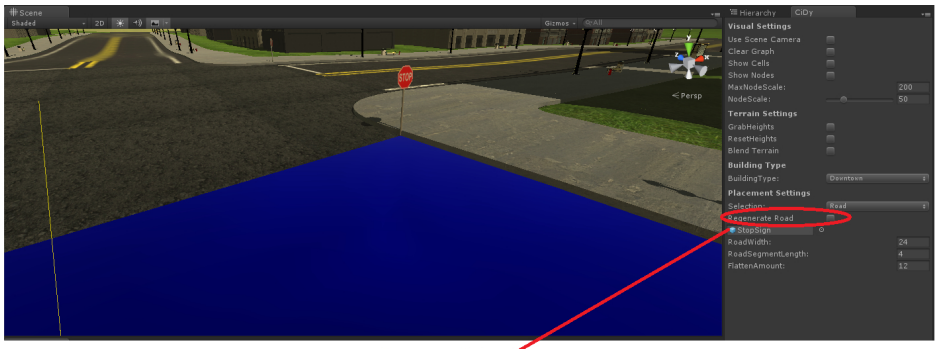
Control plus left mouse button will destroy a road. Cntrl+LMB



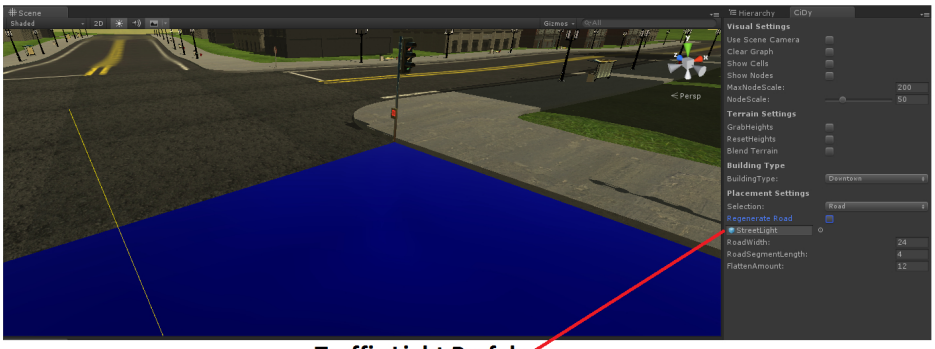
With a road selected you can press and hold right mouse button near manipulator points to modify the road path.

Any changes in values for the selected road will only take effect after pressing regenerate road.

Stop Sign GameObject : This is the prefab that will be placed at the last point of the road on the right hand side.(US Driving System)



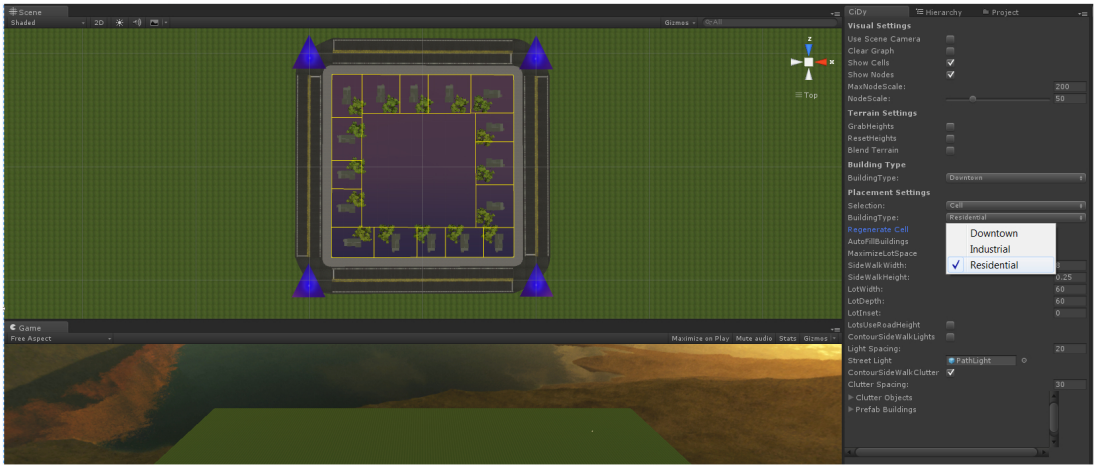
Stop Sign Prefab



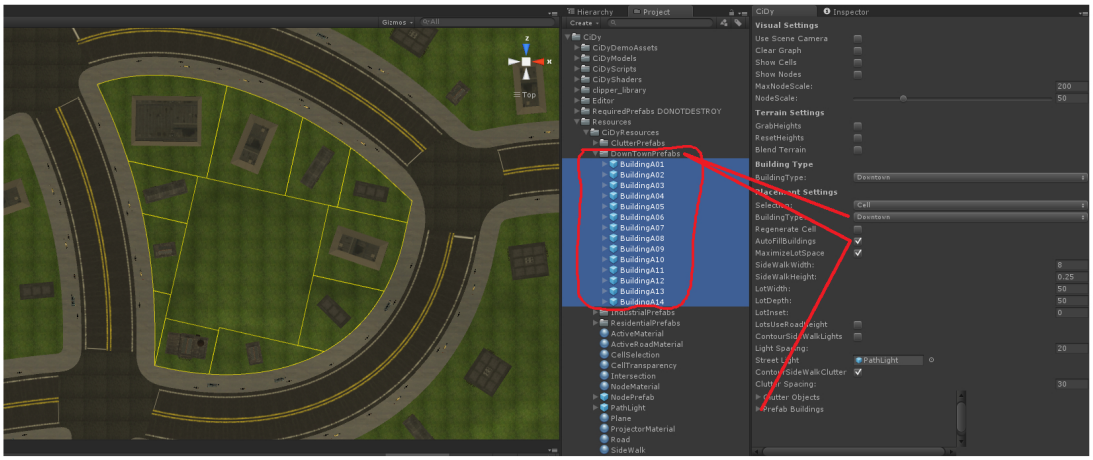
Traffic Light Prefab

Now lets look at Cell Selection.

Building Type Enum : This enum is used to determine what this cell buildingType will be set to on regenerate.(Only takes effect after pressing Regenerate Cell)



Auto Fill Buildings : This will auto fill this cells prefab buildings based on the BuildingType naming system in the CiDyResources/DowntownPrefab or IndustrialPrefabs or Residential Prefabs.



Maximize Lot Space : This is used when placing from the prefab list. If maximize is active the building placement system will pick the prefab that takes up the most space inside the lot.

Not Maximized

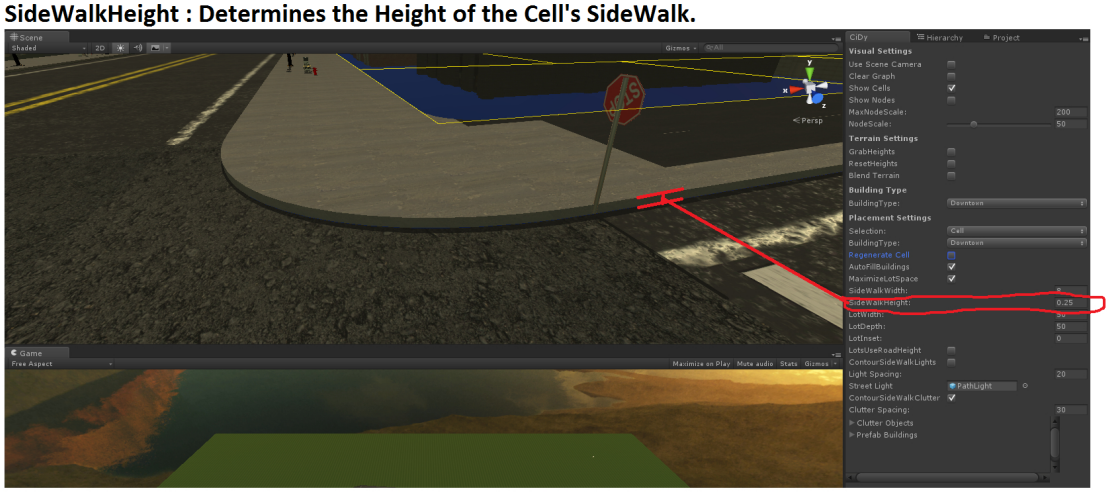
Maximized

Side Walk Width : Determines the Width of the Cell's SideWalk.

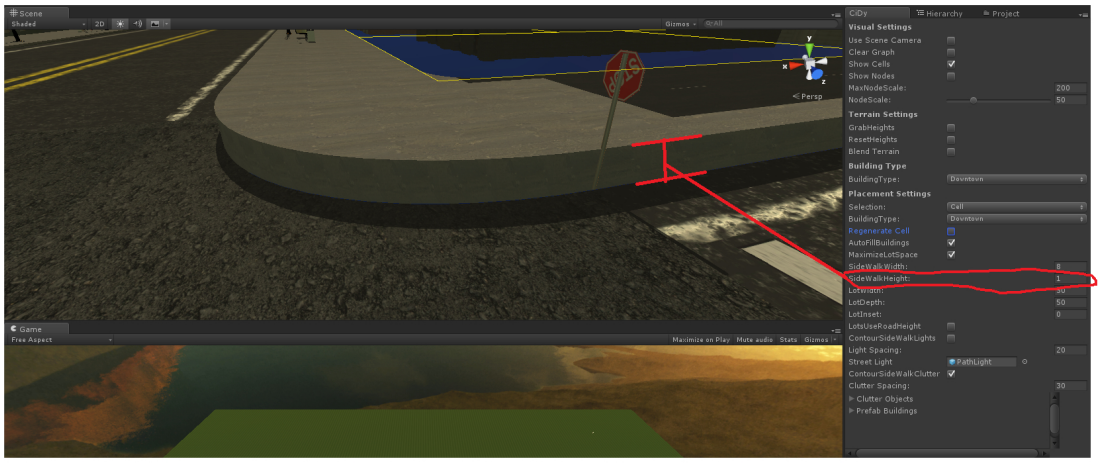
SideWalkWidth = 6

SideWalkWidth = 8

SideWalkHeight : Determines the Height of the Cell's SideWalk.



SideWalk Height = 0.25f



SideWalk Height = 1

Lot Width : Determines the desired Lot Width when subdividing cells for building placement.
Lot Depth : Determiens the desired Lot Depth when subdividing cells for building placement.

Lot Width & Lot Depth = 80

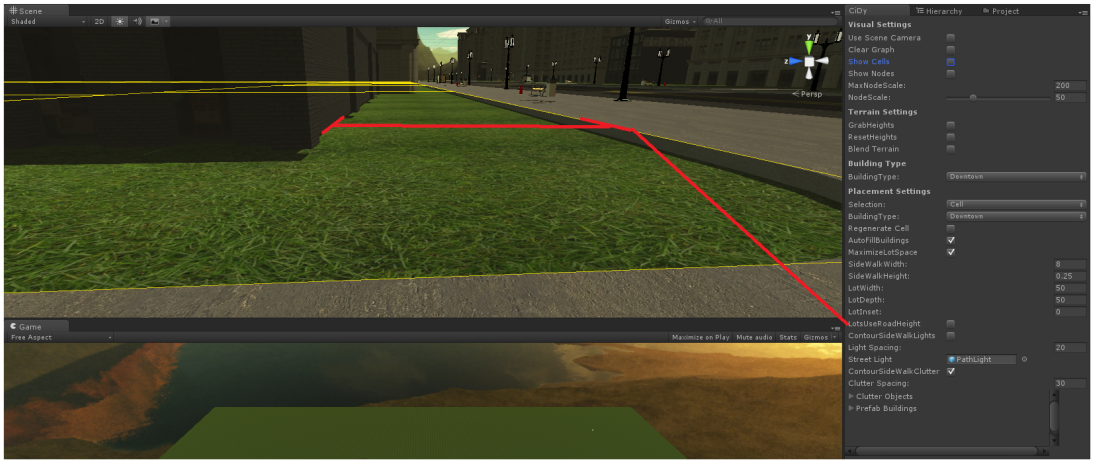
Lot Width & Lot Depth = 50

Lot Inset : Determines amount the lots will inset from eachother and sidewalks.

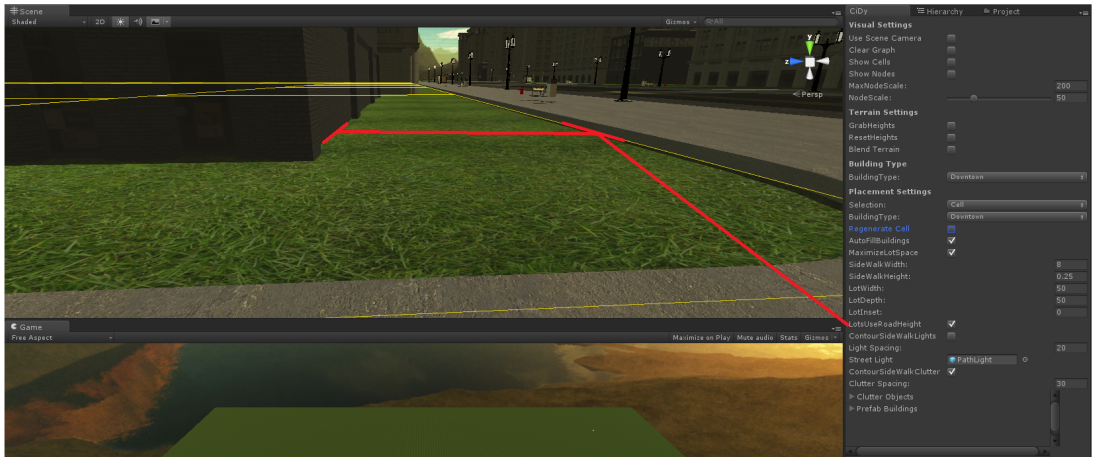
Lot Inset = 0

Lot Inset = 3

Lots Use Road Height : Determines if the cells lots heights will match the road height or the sidewalk heights. (Default if off = SideWalk Height)



LotsUseRoadHeight = off (Use SideWalk Height)



LotsUseRoadHeight = On (Use Road Height)

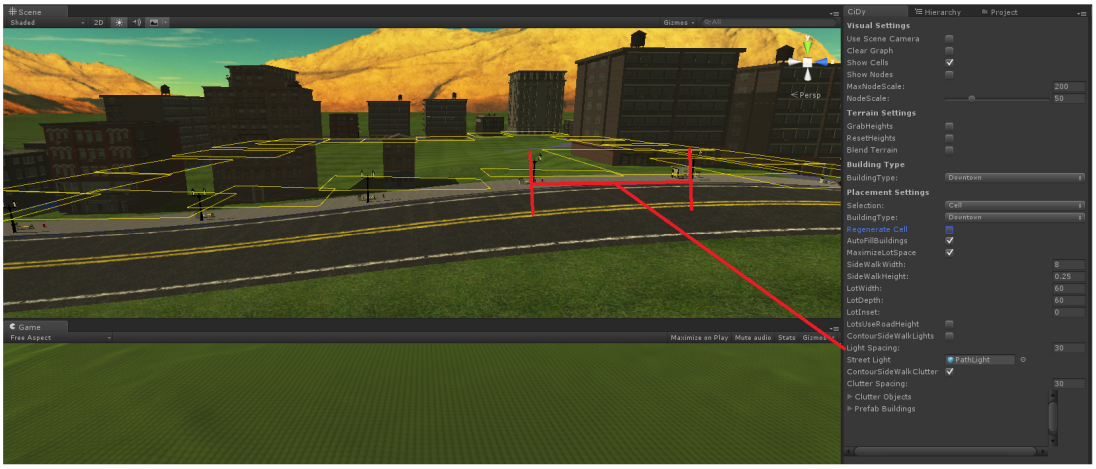
Contour SideWalk Lights : If Active prefabs will use the sidewalks local Up Vector. If Not active prefabs will use World Up Vector.

ContourSideWalkLights = off (Use World Up Vector)

ContourSideWalkLights = on (Use Local Up Vector)

Light Spacing : Determines the desired distance between each prefab placement.

Light Spacing = 10



Light Spacing = 30

Street Light GameObject : This is the prefab that will be instantiated at street light points.

ContourSideWalkClutter : If Active clutter prefabs will use SideWalk Up Vector. If not active clutter prefabs will use World Up Vector.

Clutter Spacing : Same as Light Spacing. Determines the distance between clutter points.

Clutter Objects List : The prefabs in this list will be spawned at the clutter points.

Prefab Buildings List : The prefabs in this list will be spawned and fitted into the lots. (Prefabs are rotated towards lots road acces using Forward Vector of prefab transform)